

Research Report

Statistical Analysis of Background Radiation using Geiger-Müller (GM) Counter

Abstract

An experiment was conducted to investigate the statistical nature of background ionizing radiation present in the environment using a Geiger-Müller (GM) counter. Background counts were recorded over a series of 75 independent trials with a fixed time interval of 60 seconds. The objective of the study was to determine if the fluctuations in count rate follow a Poisson distribution, which describes discrete, independent random events. Analysis of the experimental data revealed a mean count rate (μ) of 25.59 counts per min (cpm) and a variance (s^2) of 26.00. The variance-to-mean ratio was found to be 1.016, providing strong experimental evidence that background radiation decay is a random process governed by Poisson statistics.

Introduction

Radioactive decay is a fundamentally random process. Although the precise timing information regarding decay of a specific nucleus is impossible, but the probability of decay within a group of atoms is predictable over time. For random events with a constant average rate of occurrence, the data is expected to follow a Poisson Distribution. A unique characteristic of this distribution is that the variance (s^2) of the data set is approximately equal to its mean (μ):

$$s^2 \approx \mu$$

If the counts are high, typically with a mean ($\mu > 20$), the Poisson distribution approximates a Gaussian (Normal) distribution with a standard deviation (σ) = $\sqrt{\mu}$. The present experiment aimed to verify this statistical property using background radiation, which may arise from cosmic rays and terrestrial radioisotopes.

Apparatus and Methodology

The experiment was conducted using a Geiger-Müller (GM) Counter tube and scaler unit (Triode, India) with inbuilt timer. The GM tube was set up in an isolated environment with no specific radioactive source nearby, measuring only background radiation present in the environment. The GM tube was operated at a fixed voltage of 750 V; chosen specifically to lie within the 'Plateau Region' of the tube's characteristic curve, ensuring that the count rate remains stable and independent of minor voltage fluctuations. The counting interval was fixed

at 60 seconds. The counter was started, allowed to run for the fixed time interval, and the total count was recorded (Fig. 1). This process was repeated for 75 trials to ensure statistical significance. Environmental conditions were kept stable to minimize systematic errors.

Data Observations

- **Total number of trials (n):** 75
- **Time interval per trial:** 60 seconds
- **Data summary:** (See Appendix A for full data table)
 - Minimum count recorded: 10
 - Maximum count recorded: 36



Figure 1: Recording of observations on background radiation using GM Counter Apparatus.

Analysis and Calculation

The statistical parameters were calculated using the recorded dataset as described below.

(i) Mean Count Rate (μ): The average number of counts per minute was calculated as:

$$\mu = \frac{\sum_{i=1}^n x_i}{n} = 25.59 \text{ counts/min}$$

(ii) Variance (s^2): The variance, representing the spread of the data, was calculated as:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{(n - 1)} = 26.00$$

(iii) Verification of Poisson Nature: To verify the hypothesis of Poisson nature of background radiation, the ratio of variance to the mean (called the Fano Factor) was examined as:

$$\frac{s^2}{\mu} = \frac{26.00}{25.59} = 1.016$$

(iv) Graphical Analysis: A histogram of the experimental frequency was plotted against the theoretical Poisson probability curve with mean (μ) equal to $25.59 \approx 25.6$ (Fig. 2). See Appendix B for the Python code.

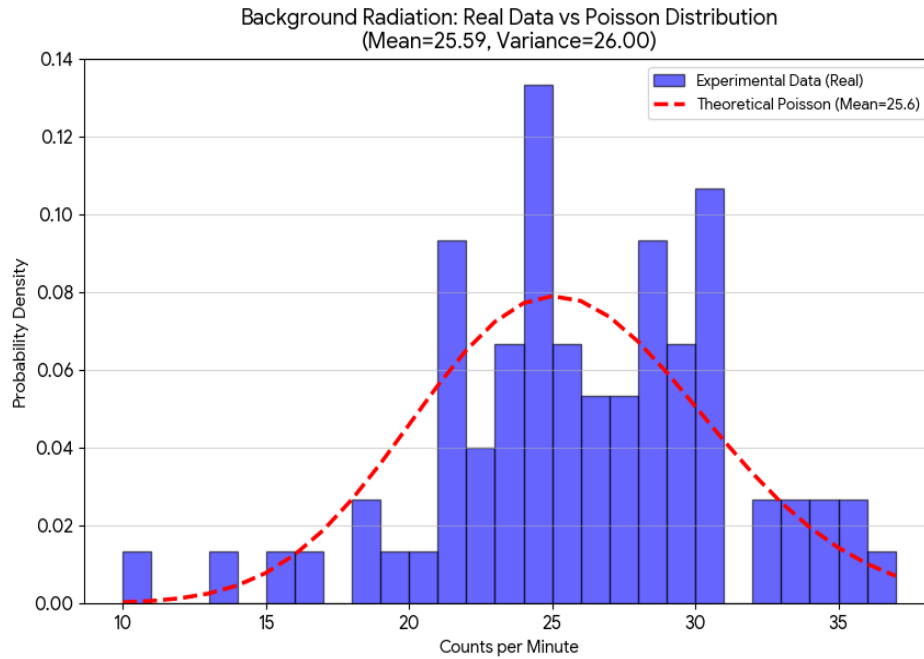


Figure 2: Comparison of Experimental Data (Bars) vs. Theoretical Poisson Distribution (Dashed Line).

Discussion

The theoretical expectation for a purely random process is that the Variance/Mean ratio equals 1. Our experimental ratio of 1.016 is exceptionally close to unity. The standard deviation of the dataset is $\sigma = \sqrt{26.00} \approx 5.10$. This confirms the "Square Root Law" where the expected error in a single radiation measurement is roughly the square root of the mean count ($\sqrt{25.59} \approx 5.06$). Visual inspection of *Figure 2* confirms that the spread in the experimental data matches the theoretical bell curve of the Poisson distribution.

Sources of Error

While the results are consistent, minor discrepancies can arise from:

- **Dead Time:** It is the finite time the GM tube takes to recover after a pulse (usually microseconds). At low background rates (~ 25 cpm), this effect is expected to be negligible.
- **Finite Sampling:** With infinite trials, the curve would be perfectly smooth. With 75 trials, small statistical bumps are expected.

Conclusion

The experiment successfully verified the statistical nature of background radiation. The data demonstrated a variance nearly close to the mean (Ratio = 1.016), confirming that radioactive decay follows Poisson Statistics. This implies that the fluctuations observed in the GM counter readings are not instrumental errors, but intrinsic physical fluctuations inherent to quantum processes.

References

1. Knoll, G. F. (2010). *Radiation Detection and Measurement*.
2. Data Source: Experimental readings taken at EMN lab, PAU Ludhiana with GM Counter on 16 July 2025.

Appendix A

Operating Voltage of GM Tube = 750 Volt	
Counting Time-interval per reading = 60 seconds	
Sr. No.	Background Count (x_i)
1	30
2	29
3	27
4	21
5	21
6	34
7	30
8	22
9	24
10	33
11	19
12	23
13	24
14	24
15	28
16	28
17	28
18	10
19	22
20	35
21	23
22	27
23	26
24	30
25	28
26	29
27	29
28	18
29	26
30	21
31	28
32	24
33	25
34	24
35	28
36	27
37	32

38	25
39	33
40	16
41	29
42	24
43	24
44	25
45	30
46	13
47	26
48	24
49	18
50	32
51	36
52	25
53	35
54	34
55	21
56	21
57	20
58	23
59	30
60	24
61	29
62	24
63	25
64	30
65	23
66	27
67	28
68	15
69	21
70	30
71	22
72	26
73	23
74	30
75	21

Appendix B

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import poisson

# 1. Load your data
# Assuming the file is in the same folder
df = pd.read_csv('GM_Counter.xlsx - Sheet1.csv', skiprows=2)
counts = df['Background Count']

# 2. Calculate Statistics
mean_val = counts.mean()
variance_val = counts.var()

print(f"Experimental Mean: {mean_val:.2f}")
print(f"Experimental Variance: {variance_val:.2f}")
print(f"Variance/Mean Ratio: {variance_val/mean_val:.2f}")

# 3. Simulation & Plotting
plt.figure(figsize=(10, 6))

# A. Plot Histogram of Your Real Data
# bins range from min count to max count
plt.hist(counts, bins=range(int(counts.min()), int(counts.max()) + 2),
         density=True, alpha=0.6, color='skyblue', edgecolor='black',
         label='Experimental Data')

# B. Plot Theoretical Poisson Curve (The Simulation)
x = np.arange(int(counts.min()), int(counts.max()) + 2)
pmf = poisson.pmf(x, mean_val) # Probability Mass Function
plt.plot(x, pmf, 'r--', linewidth=3, label=f'Theoretical Poisson (Mean={mean_val:.1f})')

# Labels
plt.title(f'Poisson Verification: Background Radiation\n(Mean={mean_val:.2f}, Var={variance_val:.2f})')
plt.xlabel('Counts per Minute')
plt.ylabel('Probability')
plt.legend()
plt.grid(axis='y', alpha=0.3)

plt.show()
```